

BREMERTON, WA, USAWMO: **727928**Lat: **47.483N**Lon: **122.767W**Elev: **444**StdP: **14.46**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **94263****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.8	26.7	12.2	10.4	29.1	18.0	13.8	30.8	24.0	45.0	20.4	44.2	2.9	40	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	86.4	65.6	82.1	63.9	79.1	62.5	66.9	83.4	65.2	80.4	63.4	77.1	7.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	81.2	71.6	58.8	75.2	69.5	57.2	70.8	67.7	31.5	83.8	30.2	80.5	28.9	77.1	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.2	15.3	DB	16.7	93.2	5.2	4.2	12.9	96.2	9.9	98.7	7.0	101.0	3.2	104.0	(4)
(5)			WB	15.7	69.5	4.8	2.0	12.3	70.9	9.5	72.0	6.9	73.1	3.4	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	39.3	40.3	43.7	47.5	53.7	58.4	63.7	63.8	58.9	50.0	43.1	38.2	(6)	
	DBStd	10.48	5.71	5.15	5.19	5.50	5.63	5.29	5.02	4.48	4.95	4.90	5.56	5.71	(7)	
	HDD50	1590	332	272	205	111	24	1	0	0	2	62	214	367	(8)	
	HDD65	5542	796	691	660	525	354	212	84	76	191	466	656	831	(9)	
	CDD50	1634	1	1	11	36	137	254	425	429	269	61	9	1	(10)	
	CDD65	109	0	0	0	0	2	14	44	40	9	0	0	0	(11)	
	CDH74	2064	0	0	3	19	85	270	782	694	210	1	0	0	(12)	
	CDH80	623	0	0	0	2	12	84	272	207	46	0	0	0	(13)	
(14) Wind	WSAvg	5.6	5.7	5.9	6.6	6.1	5.7	5.7	5.2	4.9	5.2	5.5	5.7	5.5	(14)	
(15) Precipitation	PrecAvg	52.8	8.1	5.6	5.8	3.7	2.3	1.7	0.7	1.1	1.9	5.2	8.4	8.3	(15)	
	PrecMax	66.6	16.8	14.4	12.2	8.5	4.6	3.4	2.0	3.7	8.0	12.3	19.9	14.6	(16)	
	PrecMin	39.2	3.8	0.2	1.4	1.2	0.2	0.2	0.1	0.0	0.1	0.6	2.4	2.8	(17)	
	PrecStd	8.7	2.9	2.8	2.4	1.9	1.2	0.9	0.6	1.0	1.7	2.8	4.0	3.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	57.4	69.0	76.7	81.6	88.5	92.9	90.1	84.4	72.0	60.1	54.2	(19)	
		MCWB	50.7	47.5	53.5	57.7	61.2	65.3	68.7	66.3	63.9	57.6	55.0	52.2	(20)	
	2%	DB	52.1	54.1	61.1	68.2	75.4	81.9	87.7	84.8	80.6	65.7	55.4	51.5	(21)	
		MCWB	49.1	47.5	51.2	54.0	59.8	63.4	66.5	65.4	62.2	55.1	52.2	49.1	(22)	
	5%	DB	49.6	51.7	55.4	63.2	71.7	75.3	81.9	81.6	74.9	62.0	54.2	48.1	(23)	
		MCWB	47.1	47.1	48.0	51.7	57.8	60.8	64.4	64.0	60.7	54.1	51.5	46.2	(24)	
	10%	DB	47.6	48.2	52.7	57.5	65.8	71.5	78.4	77.5	71.6	58.6	52.2	46.1	(25)	
		MCWB	45.7	44.9	46.9	48.6	55.3	58.8	62.7	62.4	58.9	52.8	49.7	44.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.5	57.0	59.5	63.4	67.5	69.8	68.6	66.3	61.2	56.4	52.9	(27)	
		MCDB	54.1	53.8	61.3	72.7	76.6	85.0	89.5	85.1	80.6	65.9	58.7	53.5	(28)	
	2%	WB	49.6	50.0	52.5	55.2	60.8	63.9	67.3	66.5	63.8	57.7	53.9	48.7	(29)	
		MCDB	51.1	52.3	58.6	66.0	73.6	79.4	84.7	82.3	76.3	62.2	55.3	50.1	(30)	
	5%	WB	47.2	47.2	49.9	52.6	58.4	61.4	65.2	65.0	61.9	55.5	52.1	46.4	(31)	
		MCDB	48.4	49.1	54.9	60.7	69.4	74.8	81.0	79.7	72.8	60.3	53.5	47.4	(32)	
	10%	WB	45.8	45.6	47.4	50.3	55.9	59.3	63.1	63.1	59.7	53.7	49.6	44.8	(33)	
		MCDB	47.0	47.6	51.6	57.0	64.9	69.7	77.0	76.3	68.0	57.2	51.7	46.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	13.5	15.6	18.4	19.9	20.0	23.7	23.7	21.1	16.1	12.2	10.3	(35)	
		MCDBR	12.9	18.7	21.7	27.0	30.0	30.4	32.5	32.4	30.2	21.6	14.8	11.6	(36)	
	5% WB	MCWBR	9.9	13.0	13.4	15.4	16.0	15.4	15.6	15.8	15.8	13.6	11.0	9.6	(37)	
		MCWBR	10.3	13.6	19.2	23.6	26.2	27.7	30.6	29.7	26.1	18.2	10.9	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.324	0.311	0.324	0.345	0.370	0.356	0.344	0.348	0.336	0.325	0.322	0.324	(40)	
		taud	2.494	2.570	2.544	2.460	2.408	2.470	2.529	2.519	2.548	2.577	2.527	2.435	(41)	
	Ebn at Noon		241	275	287	289	284	288	290	285	279	265	240	223	(42)	
		Edh at Noon	21	23	27	33	36	34	31	30	27	23	20	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		318	592	887	1298	1643	1795	2004	1714	1250	703	372	260	(44)	
	RadStd		46	80	104	123	135	187	154	116	117	76	49	38	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page